

Murray S. Bennett

The Ohio State University
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EDUCATION

- Ph.D. Cognitive Psychology** 2023
University of Newcastle, Australia
Collective Workload: Human Performance in Competitive and Collaborative Tasks
- B.S. Psychology** (*1st Class Honours*) 2018
University of Newcastle, Australia

RESEARCH EXPERIENCE

- Postdoctoral Scholar** 2024-
The Ohio State University, Department of Psychology
- Applied and advanced quantitative cognitive models (SFT,GRT) to investigate processing independence in social and cognitive decision-making.
 - Engineered new analytical tools and trained deep neural networks to enhance human-AI collaboration for melanoma identification.
- Postdoctoral Researcher** 2022-24
University of Texas at San Antonio, Psychology Department
- Quantified and combined human perceptual judgements and computer vision to show human-AI collective benefit for melanoma prediction.
 - General Recognition Theory examination of perceptual interactions in melanoma judgements.
- Data Analyst** 2021-22
Hunter New England Health Local Health District
- Conducted weekly analysis of nationwide FluTracking and COVID-19 survey data for Australian federal government reporting.
 - Collaborated with international teams to establish survey infrastructure and data collection for New Zealand and Argentina.
- Research Assistant** 2021
University of Newcastle, School of Biomedical Sciences and Pharmacy
- Designed and conducted an experiment on the physiological correlates of cognitive workload using the Detection Response Task.
 - Provided specialised expertise and analysis for cognitive workload, cognitive performance, and physiological measures.

Research Assistant

2019-22

University of Newcastle, School of Psychology

- Provided technical and methodological expertise on research projects in cognitive processing and decision-making.
- Designed and coded online versions of foundational cognitive psychology experiments to support undergraduate coursework.

Research Internship

2018

University of Newcastle, Center for Brain and Mental Health Research

- Joint analysis of cognitive workload and flight-simulator performance of helicopter pilots testing novel 3D helmet displays.

TEACHING EXPERIENCE

Course Instructor

Introduction to Psychology

UTSA 2024

- Introduction to the study of and scientific approaches to attention, sensation, perception, emotion, learning, memory, problem solving, personality, development, atypical behaviour, and social behaviour.

Honours Thesis Grader

University of Newcastle, School of Psychology

2021

- Assessed and critiqued the methodology, data analysis, and scientific writing of honours-level cognitive psychology research projects.

Lab Demonstrator

Advanced Research Methods & Design

UoN 2021

- Presenting practical content and guiding students in the hands-on application of advanced concepts in research methodology, statistics, Bayesian analysis, and experimental design.

Cognitive Psychology

UoN 2020

- Introduction and overview of fundamental cognitive psychology principles, including attention, memory, language, reasoning, and problem solving, with focus on the application of experimental methods and formal psychology report writing.

Accreditation

Associate *Centre for the Integration of Research, Teaching and Learning*, 2024

- Associate-level teaching accreditation via CIRTl by completing professional development events and demonstrating expertise in evidence-based teaching.

ACHIEVEMENTS & AWARDS

People's Choice Best Presentation Virtual Mathematical Psychology Conference	2024
Best Poster UT Health Postdoctoral Research Forum	2023
Summer Research Internship University of Newcastle's Center for Brain and Mental Health	2019 \$2,000
Best Presentation University of Newcastle's Student Research Conference	2018
Faculty of Science Commendation List University of Newcastle	2018

ACADEMIC SERVICE

Conference session chair Society for Mathematical Psychology conference	2025
Graduate research presentations judge The Ohio State University's Hayes advanced research forum	
Founded and chaired postdoctoral writing club University of Texas at San Antonio	2024
Postdoctoral appreciation week organisational committee University of Texas at San Antonio	
Postdoctoral mentor of the year judge University of Texas at San Antonio	
Abstract review Southwestern psychological association convention	
3-minute thesis judge University of Texas at San Antonio	2023
Postdoctoral association member University of Texas at San Antonio	
Conference session chair Australian Society for Mathematical Psychology conference University of Newcastle, school of psychology's honours research conference	2022
Board of directors student representative Newcastle University Sport	2021

PROFESSIONAL SOCIETIES

Society for Mathematical Psychology	2019-present
Psychonomic Society	2019-present
Southwestern Psychological Association	2024

PUBLICATIONS

Journal Publications

11. Gronau, Q. F., **Bennett, M. S.**, Brown, S. D., Hawkins, G. E., and Eidels, A. (2023). Do choice tasks and rating scales elicit the same judgments? *Journal of Choice Modelling*, 49:100437
10. **Bennett, M. S.** (2023). Collective workload: Human performance in competitive and collaborative tasks. *NOVA repository*
9. **Bennett, M. S.**, Hedley, L. G., Love, J., Houpt, J. W., Brown, S. D., and Eidels, A. (2022). Human performance in competitive and collaborative human-machine teams. *Topics in Cognitive Science*
8. John, A. R., Singh, A. K., Do, T.-T. N., Eidels, A., Nalivaiko, E., Gavgani, A. M., Brown, S. D., **Bennett, M. S.**, Lal, S., Simpson, A. M., et al. (2022). Unraveling the physiological correlates of mental workload variations in tracking and collision prediction tasks. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 30:770–781
7. Garrett, P. M., **Bennett, M. S.**, Hsieh, Y.-T., Howard, Z. L., Yang, C.-T., R., D., and Eidels, A. (2022). Wheel of fortune: A cross-cultural examination of how expertise shapes the mental representations of familiar and unfamiliar numerals. *Computational Brain & Behavior*, 5(1):45–59
6. **Bennett, M. S.**, Mullard, R., Adam, M. T., Steyvers, M., Brown, S. D., and Eidels, A. (2020). Going, going, gone: competitive decision-making in dutch auctions. *Cognitive Research: Principles and Implications*, 5(1):1–22

In Preparation or Under Review

5. **Bennett, M. S.**, Houpt, J. W., Wenger, M. J., Townsend, J. T., and Mark, L. (2025). Enhancing melanoma detection by combining continuous feature estimates from human perception and computer vision. *Preprint*, <https://osf.io/ctg9s>
4. Houpt, J. W., **Bennett, M. S.**, Silvas, E., and Chen, Y.-Y. (2025). Applied configural processing. *In preparation*
3. Hedley, L. G., **Bennett, M. S.**, Love, J., Brown, S. D., and Eidels, A. (2025a). Benchmarking the benchmark: A novel simulation benchmark for team performance. *Under review*
2. Hedley, L. G., **Bennett, M. S.**, Love, J., Brown, S. D., and Eidels, A. (2025b). Division of labour and team performance: Human-human teams compared to hybrid human-machine teams. *Under review*

Conference Proceedings

1. Hedley, L. G., **Bennett, M. S.**, Love, J., Houpt, J. W., Brown, S. D., and Eidels, A. (2023). The relationship between teaming behaviours and joint capacity of hybrid human-machine teams. In *Proceedings of the Annual Meeting of the Cognitive Science Society*, volume 45

Reviewing Activity

Ad hoc reviewer for: *Cognitive Research: Principles and Implications, Computational Brain and Behaviour, and Topics in Cognitive Science, Communications Psychology*

PRESENTATIONS

Invited Talks

- 2024
3. *Human-AI perceptual judgments of malignant melanoma*. Psychology Department, The University of Newcastle, Australia.
 2. *Human-AI collaboration*. Department of Psychological Sciences at Tarleton State University.
 1. *Collaborative and competitive 'Team Spirit'*. Joint meeting of the Newcastle Cognition Lab and UC Irvine's MADLAB.

Platform Presentations

15. **Bennett, M. S.**, Alaukik, A., Smith, C., and Kvam, P. (2025, July 25-28). Processing independence in implicit attitudes: An sft and grt analysis. In *Meeting of the Society for Mathematical Psychology/ICCM*
14. **Bennett, M. S.** and Houpt, J. W. (2024, June 17-21). A grt examination of the abcs for melanoma identification with comparison to a deep convolutional neural network. In *Virtual Meeting of the Society for Mathematical Psychology/ICCM*
13. **Bennett, M. S.** and Houpt, J. W. (2024, March 27-30). Using ai and human feature perception to improve melanoma identification. In *Southwestern Psychology Association Convention*
12. **Bennett, M. S.** and Houpt, J. W. (2023, October 6-7). Improving human perception and computer vision identification of malignant melanoma. In *The Association for Research in Memory, Attention, Decision-making, Imaging, Language, Learning, and Organization*
11. **Bennett, M. S.** and Houpt, J. W. (2023, July 18-21). A bradley-terry-luce model-based analysis of human perception and computer vision to identify melanoma lesions. In *Meeting of the Society for Mathematical Psychology*
10. **Bennett, M. S.** (2021, April 9-11). Cognitive workload and performance of collaborative and competitive teams. In *Australasian Experimental Psychology Society*
9. **Bennett, M. S.** (2021, February 11-12). Cognitive workload and performance of collaborative and competitive teams. In *Australasian Mathematical Psychology Conference, Newcastle*

8. **Bennett, M. S.** (2020, October 16). Effects of competitive and collaborative team environments on individual cognitive load. In *Centre for Brain and Mental Health Research Post-Graduate and Postdoctoral Conference, Newcastle*
7. **Bennett, M. S.** (2020, July 20-31). An iterated prospect theory model for the dutch auction. In *Society for Mathematical Psychology/ICCM Virtual Conference*
6. **Bennett, M. S.**, Mullard, R., Adam, M. T., Steyvers, M., Brown, S. D., and Eidels, A. (2020, February 12-14). Competitive decision-making in dutch auctions. In *Australasian Mathematical Psychology Conference, Sydney*
5. Eidels, A. and **Bennett, M. S.** (2019, November 13). Workload capacity in collaborative and competitive group performance. In *Configural Processing Consortium, Montréal*
4. **Bennett, M. S.** and Eidels, A. (2019, August 2). Mental workload in a team based dual task. In *Centre for Brain and Mental Health Research Post-Graduate Conference, Newcastle*
3. **Bennett, M. S.**, Eidels, A., and Garrett, P. M. (2019, February 14-16). The cost of errors: Confusion analysis and the mental representation of numerals. In *Australasian Mathematical Psychology Conference, Melbourne*
2. **Bennett, M. S.**, Garrett, P. M., Yang, C.-T., and Eidels, A. (2018, November 14). Mental representation of familiar and unfamiliar numerals. In *Configural Processing Consortium, New Orleans*
1. **Bennett, M. S.** and Eidels, A. (2018, November 9). At sixes and sevens: The mental representation of familiar and unfamiliar symbolic numerals. In *University of Newcastle Psychology Student Research Conference, Ourimbah*

Poster Presentations

3. **Bennett M., S.** (2025, November 20-23). Integrating human and ai perception for enhanced melanoma diagnosis. In . *66th Annual Meeting of the Psychonomic Society*
2. **Bennett M., S.** (2023, September 19). Improving human perception and computer vision identification of malignant melanoma. In . *San Antonio Postdoctoral Research Forum*
1. **Bennett, M. S.** (2023, September 22). Human perception of melanoma features vs computer vision algorithm. In . *University of Texas at San Antonio Postdoctoral Showcase*